

PW 4: Sequences

Goal

- Understand and know how to use sequences and their operations
- Write and test subprograms (**always!**)
- Write docstring and explanations (**always!**)

Exercise 1: Sequence operations

Write the Python instructions that achieve the goals expressed in comments below. We will use assert to indicate the result of written expressions or the effect of written instructions.

```
" 'Sequence operations (with a list).' "
```

```
# Initialize a name s with a list containing in order 9, 1, 5, 2, 1 and 3
```

```
# Get the size of the list s (the number of elements it contains)
```

```
# Find out if element 2 is present in s. Same for element 4
```

```
# Get the first element of s
```

```
# Get the last element of s
```

```
# Get the frequency (number of occurrences) of 1 in s
```

```
# Remove element at index 2 of s
```

```
# Remove element 2 from s
```

```
# Add 7 at the end of s
```

```
# Add 6 in position 2 in s
```

```
# Add 0 before the first occurrence of x in s (x could be 1, 3 ...)
```

```
# Get the sum of the squares of the numbers of s?
```

Exercise 2: List

Indicate the effect of the instructions of the following program:

```
s = []
```

```
s.append(2)          # s ?
```

```
s.insert(0, 4)       # s ?
```

```
s.insert(20, 7)      # s ?
```

```
s[1] = 'd'          # s ?
```

```
s[2] /= s[2]         # s ?
```

```
s.count(1)           # ?
```

```
s[0], s[1] = s[1], s[0] # s ?
```

```
p, _, d = s          # p ? _ ? d ?
```

```
first, *following = s # first ? following ?
```

```
b = [False, True]
```

```
s.extend(b)          # s ?
```

```
x = s.pop(1)          # x ? s ?
```

```
s2 = [2, 3, 5]
```

```
i, s2[i], x = s2       # i ? s2 ? x ?
```

```
s.append(s2)           # s ?
```

```
s2.append(s)           # s2 ? s ?
```

```
t = tuple(b)           # t ?
```

```
s = list('End.')        # s ? s2 ?
```

Exercise 3: Answer the following questions concisely and precisely.

- 3.1. Explain what an immutable object is. Give examples
- 3.2. What are the sequences offered by Python? Give an example of each.
- 3.3. Indicate what “is” and “==” mean in Python. Give an example.

Exercise 4: tuple

Consider the tuple `t = (5, 'x', [])`.

- 4.1. Indicate how to access the components of the tuple: 5, 'x' and [].
- 4.2. Indicate whether the following instructions are allowed and, if so, what is t.

```
t[0] = 6
```

```
del t[1]
```

```
t[2].append(-1)
```

Exercise 5: Vowels and consonants

Write a subprogram that indicates whether a character is a consonant, vowel, or other character.

Exercise 6: Number of days in a month

Write a subprogram that calculates the number of days in a month.

Exercise 7: Operations on sequences

The `operations_sequences.py` file gives the specification of some operations inspired by those found on Python sequences as well as programs for testing these operations. Write the codes of these operations and use the test programs to test them. Warning: It is forbidden to use `break` or `return` in a repetition.